Campus Parking and Exit Carpool Coordination

Final Report

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Executive Summary

This report presents the complete systems engineering process and final design for the Campus Parking and Exit Carpool Coordination project, a socio-technical system intended to manage parking capacity and coordinate shared trips at the engineering campus of Universidad Distrital. The proposal evolved across four workshops. Workshop 1 produced an empirical diagnosis of campus mobility through field observations, occupancy measurements, process documentation, and a survey of 51 users. Workshop 2 defined an initial architecture centered on reservation management, parking status monitoring, and exit-time carpool coordination. Workshop 3 reframed the initiative as a robust socio-technical system, introducing a refined architecture, a risk and quality framework aligned with ISO 31000 and ISO/IEC 25010, and a project management plan. Workshop 4 translated this design into two complementary simulations: a process-oriented model and a behavior-oriented model to validate system behavior under baseline, optimization, and failure scenarios.

Empirical data show that more than half of users arrive before 8:00 a.m., that daily overflow reaches approximately 90 motorcycles, and that 34.3% of respondents initially express willingness to use carpooling. The conceptual design addresses these conditions through coordinated parking allocation, rule-based prioritization for shared vehicles, and exit-time matching by destination zones, supported by accountability and trust mechanisms.
Simulation results confirm that the proposed system is viable only if certain critical conditions are protected. With current capacity (110 spaces), overflow cannot be eliminated without high carpool matching performance; institutional incentives are required for adoption to grow beyond its baseline; and synchronization failures combined with cancellations can trigger nonlinear trust collapse. Under optimized conditions, higher match rates and more reliable operation

significantly reduce overflow and strengthen trust and adoption, whereas failure conditions push the system back toward unmanaged parking and informal coordination.

The report concludes that the project is ready for a controlled pilot, provided that the university treats reliability, incentives, accountability, and high-risk operating conditions as first order design concerns rather than secondary adjustments. Recommendations include a rainy-Friday response protocol, explicit trust recovery mechanisms, refinement of participation rules to overcome adoption ceilings, and continuous monitoring of key performance and behavioral indicators.

1. Introduction

Urban university campuses face strong constraints on parking, safety, and sustainable mobility, particularly when physical capacity is fixed and demand is highly concentrated in a few arrival windows. At the engineering faculty of Universidad Distrital, field observations revealed recurrent saturation of the Calle 40 basement parking area and overflow into surrounding public streets, creating operational, safety, and governance challenges.

The Campus Parking and Exit Carpool Coordination project addresses these challenges by designing a socio-technical coordination system, not merely an application. The system aims to (i) manage parking inventory through reservations and real-time status, (ii) prioritize shared vehicles through explicit institutional rules, (iii) coordinate exit-time carpool trips from campus to key zones across the city, and (iv) provide oversight and accountability mechanisms for guards and administrators.

From the beginning, the project adopted a systems-thinking perspective: technology is treated as one enabling component within a larger mobility ecosystem involving students, security staff, campus administration, transportation alternatives, and institutional policy. This report integrates the analytical, architectural, managerial, and simulation results into a final, coherent view of the system.

2. Project Evolution and Problem Understanding

2.1 Workshop 1 – Empirical Diagnosis

Workshop 1 focused on system analysis through primary data collection and field work. The team conducted parking occupancy measurements, process observations at the Calle 40 basement entrance, and a structured survey administered to 51 community members. Key findings included:

- Around 55% of respondents arrive before 8:00 a.m., creating a recurrent early-morning saturation peak.
- Approximately 90 motorcycles overflow onto surrounding public streets on high-demand days.
- Access control depends on a mix of digital registration and manual guard decisions, producing bottlenecks and inconsistency.
- Informal ride-sharing via WhatsApp and other messaging apps is already widespread, driven by cost, convenience, and safety reasons.

These observations established the core problem: parking is being managed reactively and locally at the gate, while coordination of arrivals and shared trips remains fragmented and invisible to the institution.

2.2 Workshop 2 – Initial Architecture

Workshop 2 translated these findings into an initial architecture that included a mobile/web application for students, a dashboard for security staff, ride-matching and status-update services, and a data and monitoring layer. At that stage, the project still reflected the original Sustainable Transportation Coordinator concept, which combined inter-campus carpooling and an institutional bike-sharing system.

The architecture already incorporated requirements for scalability, reliability, and data privacy, but the scope remained ambitious and partially misaligned with the changing institutional context, particularly the end of the ECCI campus contract. This realization triggered a deeper feasibility review and scope refinement.

2.3 Workshop 3 – Scope Refinement and Robust Design

Between Workshops 2 and 3, the team reassessed the project's feasibility. The analysis concluded that a short-term institutional bike-sharing system was not realistic due to infrastructure, safety, and operational constraints at the Calle 40 campus. Additionally, the inter-campus mobility focus lost priority when the ECCI contract was not renewed.

As a result, the project was renamed and refocused as Campus Parking and Exit Carpool Coordinator, concentrating on:

- Reservation, allocation, supervision, and release of parking spaces.
- Policy-based prioritization for shared vehicles.
- Exit-time carpool coordination from campus to destination zones across the city.

Workshop 3 deliberately shifted from describing a future software product to defining a robust socio-technical system that integrates operational rules, stakeholder roles, physical constraints, risk management, quality attributes, and implementation planning.

2.4 Workshop 4 – Simulation and Validation

Workshop 4 operationalized the conceptual design through computational simulation and validation. The team developed:

- A Process-Oriented Simulation (discrete-event) for operational flows—parking demand, overflow, carpool requests, cancellations, synchronization failures, and matching outcomes.
- A Behavior-Oriented Simulation (agent-based) for long-term dynamics—adoption, trust, WhatsApp dependence, parking saturation, driver ratio, and feedback loops.

Three scenarios—baseline, optimization, and failure—were defined to test the system under normal, improved, and degraded conditions. This stage directly addresses the professor's request for stronger analytical support, capacity reasoning, and empirical validation of design decisions.

3. Conceptual System Design

3.1 Stakeholders and Socio-Technical Perspective

The stakeholder analysis identifies students, faculty and staff, security guards, campus administration, IT, University Welfare, and environmental management units as key actors, each with specific needs, interactions, benefits, and potential frictions. Students and faculty require predictable parking and, for some, opportunities to participate in shared trips; guards require consistent, data-informed access decisions; administrators require traceability and policy enforcement; IT provides authentication and infrastructure; and welfare and environmental units consume mobility data for quality-of-life and sustainability initiatives.

An important design decision is to treat stakeholders as interacting with policies, physical constraints, and decision procedures, not just with interfaces. This preserves the distinction between the broader socio-technical system and the supporting platform, aligning with the course's emphasis on systems engineering rather than pure software development.

3.2 Problem Modelling

3.2.1 Context Diagram

The context diagram defines the system boundary by identifying external entities—students, guards, campus administration, and the institutional authentication service—and the flows of information between them. Students submit reservation and carpool requests; guards receive access approvals and occupancy alerts and update entry/exit logs; administrators configure policies and consume reports; and the authentication system validates institutional credentials.

This model clarifies that the system's boundary includes the campus parking area and internal coordination processes, while the surrounding public streets remain an external environment where the consequences of poor coordination become visible.

3.2.2 As-Is vs. To-Be Process Flow

The As-Is flow reveals two fundamental weaknesses: access decisions depend on guards' visual judgment and there is no structured outcome when the basement is full—students are implicitly redirected to public streets without any record or notification. The To-Be flow shifts decision-making from reactive, on-site evaluation to proactive digital reservation and data-driven allocation, generating explicit notifications for both successful assignments and waiting-list placements.

This transformation from manual, gate-centered control to coordinated, pre-arrival management is the central operational improvement of the system.

3.2.3 Input–Process–Output Model

The IPO model identifies four main input–process–output chains: reservation and authentication leading to assigned spaces, vehicle data leading to availability confirmation, carpool requests leading to confirmed shared routes, and credential-based policy application leading to saturation alerts. The model emphasizes that the system's value lies in transforming fragmented informal inputs into structured, actionable outputs for each actor, rather than simply storing data.

3.3 Architecture Overview and Subsystems

The conceptual architecture organizes the system into four core components: student participation, parking state management, exit coordination, and operational control. Student participation introduces requests and updates; parking state management maintains the status of spaces; exit coordination matches drivers and passengers; and operational control supervises compliance, exceptions, and interventions.

These components are further decomposed into functional subsystems—access control, parking state, allocation and policy, exit coordination, and information update—each responsible for a specific operational need such as institutional legitimacy, occupancy consistency, rule enforcement, route matching, and notification delivery.

Information flows connect these subsystems: user inputs change system state; allocation decisions depend on real-time parking status and policies; matching outcomes affect later departures; and operational feedback loop closes through guard interventions and administrative oversight. This decomposition maintains conceptual clarity while avoiding premature commitment to a particular technology stack, directly addressing the instructor's emphasis on system-level design.

4. Risk, Quality, and Project Management

4.1 Risk Identification and Categorization

The risk framework recognizes both technical and human dimensions, since the system operates as a socio-technical arrangement. Key risks include:

- Low user adoption and persistence of informal WhatsApp coordination (R1).

- Cancellation, no-show, and repeated misuse behaviors undermining trust and credibility (R2).
- Parking-state desynchronization between physical reality and digital records (R3).
- Institutional misalignment or lack of administrative support (R4).
- Scalability limits as more users or zones are incorporated (R5).

Each risk is classified according to likelihood and impact, and mapped to specific system components and project activities, strengthening traceability between threats, mitigation strategies, and design decisions.

4.2 Accountability, Trust, and Participation Mechanisms

Given the role of cancellations, no-shows, and misuse in the failure dynamics identified by Workshop 4, the design explicitly incorporates participation accountability and trust mechanisms. Relevant behaviors include repeated late cancellations, failure to appear after reserving a space, misuse of priority spots, and inaccurate vehicle or departure information. Operational signals—reservation requests, confirmations, cancellations, check-ins, no-shows, manual overrides, and reported misuse—are recorded to differentiate isolated incidents from patterns. Graduated, rule-based consequences (warnings, temporary restrictions, loss of priority, or administrative review) aim to preserve fairness while avoiding arbitrary or disproportionate sanctions. By anchoring these mechanisms in explicit institutional policies rather than discretionary judgment, the system supports both individual accountability and institutional legitimacy, directly responding to the instructor's emphasis on trust and responsibility.

4.3 Quality Attributes and Metrics

The project adopts ISO 31000 for risk management and ISO/IEC 25010 for quality attributes, focusing on functional suitability, reliability, usability, performance efficiency, security, and maintainability. Quality metrics include:

- Maximum acceptable delay for reservation confirmation during peak windows.
- Upper bounds on parking state mismatch between system records and physical occupation.
- Target thresholds for cancellation and no-show rates before corrective measures are triggered.
- Limits on average intervention time for conflict resolution by guards.

These metrics connect directly to simulation parameters and pilot monitoring indicators, strengthening the linkage between analysis, design, quality, and validation requested in the feedback.

4.4 Capacity, Scalability, and Sustainability

Capacity assumptions explicitly incorporate empirical data: 110 basement spaces, overflow of approximately 90 motorcycles, and early-morning arrival clustering. Performance targets are defined for handling peak request volumes during the 7:00–8:00 and 12:00–14:00 windows, when up to several hundred users may interact with the system simultaneously.

Scalability is addressed conceptually by planning controlled expansion: increasing user volume, adding zones or parking areas, and integrating further mobility alternatives (such as future bike-sharing) are treated as phased changes with explicit review of policy, infrastructure, and system limits. The sustainability discussion recognizes that while bike-sharing is not presently feasible, the system creates a foundation for future low-emission initiatives by formalizing carpooling and providing data on vehicle reduction.

4.5 Project Management and Agile Planning

The project management approach aligns with iterative development rather than strict waterfall sequencing. Planning artifacts are interpreted as cycles of refinement across analysis, design, risk management, and validation, rather than as one-pass stages. This alignment addresses the instructor's concern that scheduling diagrams should reflect agile principles when claimed as such.

5. Simulation and Validation

5.1 Simulation Objectives

Workshop 4's simulations aim to answer whether the conceptual design remains coherent and viable under realistic operating conditions, and to identify the specific conditions under which the system fails. The simulations test:

- Whether coordinated matching can meaningfully reduce overflow given current capacity.
- Whether adoption and trust can grow without institutional support.
- How cancellations and synchronization failures interact with trust.
- How rain and high-demand days combine to create critical overload.

5.2 Simulation Paradigms

Two paradigms were selected based on the nature of the system:

- Discrete-Event Simulation for the process-oriented model, because parking and coordination states change at discrete events such as arrivals, reservation requests, allocation decisions, and cancellations.
- Agent-Based Modelling for the behavior-oriented model, because adoption and trust emerge from repeated interactions between many individuals, system reliability, incentives, and perceived usefulness.

This combination captures both operational sequences and collective behavioral dynamics, addressing the course requirement to explore complexity and emergent behavior.

5.3 Process-Oriented Simulation

The process-oriented model simulates a seven-day cycle with day-specific motorcycle demand, a fixed capacity of 110 spaces, rain events, overflow generation, carpool request ranges, cancellations, synchronization failures, and matching outcomes. Demand ranges and overflow thresholds are calibrated from Workshop 1 observations; scenario-specific match, failure, and cancellation rates translate the risk framework from Workshop 3.

The model confirms that under baseline conditions, parking demand exceeds capacity on every weekday, with overflow typically between 50 and 80 vehicles and a peak on Friday. Carpool matches remain limited due to the 45% match rate, and rain events amplify demand by a factor of 1.25, worsening overflow on rainy days.

In the optimization scenario, higher match rates and lower failure rates significantly increase successful coordination, reducing individual arrivals and thus overflow pressure. In the failure scenario, lower match rates and higher cancellations and sync issues erase these benefits and push the system back toward unmanaged conditions.

5.4 Behavior-Oriented Simulation

The behavior-oriented model simulates 500 students over 60 operational days, starting from 30% adoption, a trust score of 50, and a 15% driver ratio among active users. Each day, parking saturation, matching success, cancellations, and sync failures update adoption, trust, and WhatsApp usage through feedback rules that represent reinforcement and degradation mechanisms.

Under baseline conditions, adoption remains near its initial level, trust fluctuates around the starting value, and WhatsApp remains the dominant coordination channel. Under optimized conditions, incentives and reliability drive adoption toward the upper boundary (around 37–40%), trust grows steadily, and informal coordination declines. Under failure conditions, adoption falls toward the lower boundary (around 15%), trust collapses, and WhatsApp reclaims dominance.

5.5 Emergent Behavior and Complexity

The simulations reveal several emergent behaviors:

- Parking saturation as adoption driver: adoption increases when saturation exceeds roughly 90% and declines when it falls below about 60%, indicating that perceived necessity shapes participation.
- Trust degradation cascades: cancellations and sync failures reinforce each other, making trust loss nonlinear; once trust falls, the system loses its ability to self-recover through successful matches.
- Rain–Friday interaction: rainy Fridays generate demand far beyond typical patterns, justifying a dedicated rainy-Friday protocol rather than treating rain as a minor disturbance.
- Adoption ceiling and structural bottleneck: adoption saturates near the 40% upper boundary under current driver ratio and vehicle capacity assumptions, suggesting that incentives alone cannot overcome participation constraints.

These behaviors respond directly to the instructor's request to explore complexity, nonlinearities, and system limits rather than only average-case performance.

5.6 Model Validation, Assumptions, and Limitations

The models are single-run stochastic simulations whose parameters capture demand ranges, rain probability, match rates, failure rates, and behavioral updates. Validation relies on:

- Directional and boundary consistency across scenarios.
- Cross-model coherence between process and behavior results.
- Conceptual traceability to Workshops 1–3.

Assumptions include day-of-week demand ranges instead of full probability distributions, simplified incentive representations, and aggregated trust dynamics. Not all conceptual elements—such as full accountability escalation or detailed manual intervention processes—are parameterized as separate state transitions in the current code. The report acknowledges that repeated runs and confidence intervals are needed for formal statistical validation and identifies this as future work.

6. Discussion: Sustainable Mobility and Alternatives

Although the final scope centers on parking management and carpool coordination, the project was originally conceived as a broader Sustainable Transportation Coordinator including bike-sharing and inter-campus mobility. Feasibility analysis concluded that institutional bike-sharing is not viable in the short term, but survey data show substantial willingness (74.5%) to adopt such alternatives, indicating latent demand for lower-emission options.

The system therefore positions carpooling as a first step in a longer sustainability trajectory. By reducing individual vehicle arrivals and collecting detailed data about occupancy, shared trips, and demand patterns, the platform can support future decisions about transit integration, micro-mobility, or phased bike-sharing pilots when infrastructure and safety conditions are more favorable.

This broader view responds to the professor's recommendation to expand the analysis of transportation coordination beyond parking inventory and to examine the interaction with sustainable mobility alternatives and long-term institutional strategies.

7. Conclusions

The Campus Parking and Exit Carpool Coordination project demonstrates how systems analysis, robust conceptual design, and simulation-based validation can be combined to address a real campus mobility problem. The project has evolved from a broad sustainable mobility idea into a focused, feasible, and institutionally grounded socio-technical coordination system.

The conceptual architecture is now robust, with clearly defined components, subsystems, processes, and risk and quality frameworks. The simulations provide empirical support for key design decisions, exposing the conditions under which the system succeeds or fails and highlighting the importance of reliability, incentives, and accountability.

Before a pilot implementation, four design recommendations should be prioritized:

1. Rainy Friday Response Protocol – anticipate compounded demand through early communication, overflow redirection, and intensified carpool promotion.
2. Trust Recovery Mechanisms – implement explicit procedures to stabilize trust after clusters of cancellations or synchronization failures.
3. Participation Structure Review – reconsider driver incentives, driver ratios, and practical occupancy conditions to overcome structural adoption ceilings.
4. Reliability as Primary Non-Functional Requirement – treat synchronization reliability, data integrity, and infrastructure stability as top priorities, since they are directly linked to trust and adoption.

In systems terms, the main contribution of the project is to show that campus parking management is not only a capacity problem but a coordination and legitimacy problem. By formalizing the roles of stakeholders, rules, behaviors, and technical infrastructure, the design offers a realistic pathway for the university to move from reactive, gate-based control to proactive, data-informed and accountable mobility coordination.